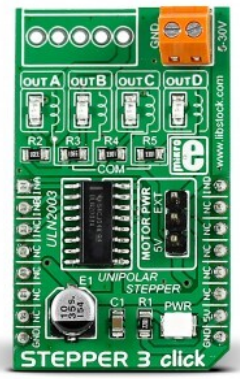
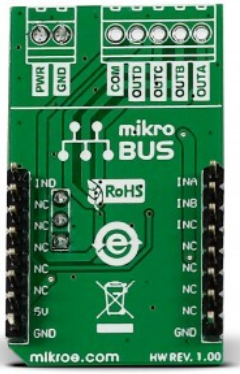
MIKROELEKTRONIKA D.O.O, Batajnički drum 23, 11000 Belgrade, Serbia
VAT: SR105917343 Registration No. 20490918
Phone: + 381 11 78 57 600 Fax: + 381 11 63 09 644 E-mail: office@mikroe.com
www.mikroe.com

Stepper 3 Click





PID: MIKROE-2035

Stepper 3 Click is a compact add-on board that contains a bipolar stepper motor driver. This board features the ULN2003A, a high-voltage, high-current Darlington transistor array from Texas Instruments. It is designed to drive unipolar stepper motors with a supply voltage applied to the common wire. The current flows through the common wire and the motor coil via the activated current sink driver to the ground. This allows driving unipolar stepper motors up to 30V and 500mA per coil. The ULN2003A itself is an ideal option for driving inductive loads such as the motor coils, as every channel is equipped with the common cathode clamp diode, which protects the IC from back EMF, typically observed at inductors and coils. This Click board™ makes the perfect solution for the development of small stepping motors in various applications such as consumer electronics and industrial equipment.

Stepper 3 click is supported by a mikroSDK compliant library, which includes functions that simplify software development. This Click board™ comes as a fully tested product, ready to be used on a system equipped with the mikroBUS™ socket.

How does it work?

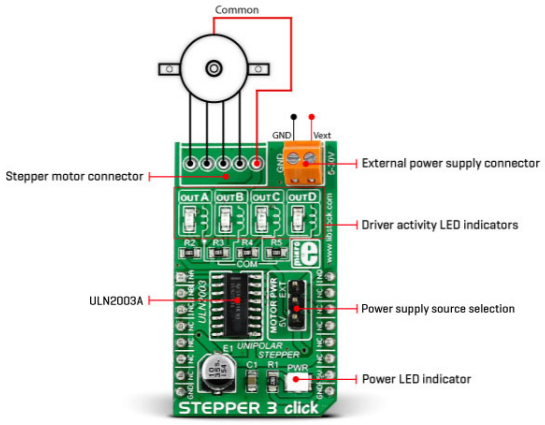
Stepper 3 Click is based on the ULN2003A, a high-voltage, high-current Darlington transistor array from Texas Instruments. Motor step progression is performed by alternating the active driver, which sinks the current through the coil connected to the respective driver. The MCU performs the alteration cycle, which controls the driver inputs: a HIGH logic level on the input pin will set the corresponding driver to a LOW logic level, allowing it to sink current. Besides driving a unipolar stepper motor as the primary function, this Click board™ can also be used for driving relays and brushed DC motors or as the logic buffer for various applications. Combining more than one driver makes it possible to sink more than 500mA.

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Stepper 3 Click uses four GPIOs to allow the host MCU to control the stepper motor or other mentioned device. Those pins are labeled as INA, INB, INC, and IND. Four output channels from the ULN2203A and a motor power supply are routed to the unpopulated 5-pin header. The motor can be powered over the 5V rail of the mikroBUSTM socket or the external power supply terminal with voltages of 5 – 30V. The selection can be made over the MOTOR PWR jumper. In addition, the ULN2003As output channels are also routed to the four output LEDs (outA, outB, outC, and outD) for a visual presentation.

This Click board™ can be operated only with a 5V logic voltage level. The board must perform appropriate logic voltage level conversion before using MCUs with different logic levels. Also, it comes equipped with a library containing functions and an example code that can be used as a reference for further development.

Specifications

Type	Stepper
Applications	Can be used for the development of unipolar stepper motor control applications. Besides, it can also be used for driving relays, brushed DC motors, etc
On-board modules	ULN2003A - high-voltage high-current Darlington transistor array from Texas Instruments
Key Features	High-power Darlington sink drivers with up to 500mA per channel, clamping diode for the kickback voltage protection, can be used for other applications such as driving relay coils or brushed DC motors, selectable power source, and more
Interface	GPIO
Feature	No ClickID
Compatibility	mikroBUS™
Click board size	M (42.9 x 25.4 mm)
Input Voltage	5V

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Pinout diagram

This table shows how the pinout on **Stepper 3 click** corresponds to the pinout on the mikroBUS™ socket (the latter shown in the two middle columns).

Notes	Pin	mikroBUS				Pin	Notes
Channel A ctrl IN	**INA**	1	AN	PWM	16	**IND**	Channel D ctrl IN
Channel B ctrl IN	**INB**	2	RST	INT	15	NC	
Channel C ctrl IN	**INC**	3	CS	RX	14	NC	
	NC	4	SCK	TX	13	NC	
	NC	5	MISO	SCL	12	NC	
	NC	6	MOSI	SDA	11	NC	
	NC	7	3.3V	5V	10	**5V**	Power supply
Ground	**GND**	8	GND	GND	9	**GND**	Ground

Onboard settings and indicators

Label	Name	Default	Description
LD1	PWR	-	Power LED indicator
LD2	outA	-	Channel A LED Indicator
LD3	outB	-	Channel B LED Indicator
LD4	outC	-	Channel C LED Indicator
LD5	outD	-	Channel D LED Indicator
J1B	MOTOR PWR	-	Motor Power Supply Source Selection 5V/EXT: Down position 35V, Up position EXT

Stepper 3 click electrical specifications

Description	Min	Typ	Max	Unit
Supply Voltage	-	5	-	V
External Power Supply Voltage	5	-	30	V
Current Limit per Channel	-	-	500	mA

Software Support

We provide a library for the Stepper 3 Click as well as a demo application (example), developed using MIKROE compilers. The demo can run on all the main MIKROE development boards.

Package can be downloaded/installed directly from NECTO Studio Package Manager(recommended), downloaded from our LibStock™ or found on Mikroe github account.

Library Description

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This library contains API for Stepper 3 Click driver.

Key functions

- stepper3_set_step_mode This function sets the step mode resolution settings in ctx->step_mode.
- stepper3_set_direction This function sets the motor direction to clockwise or counter-clockwise in ctx->direction.
- stepper3_drive_motor This function drives the motor for the specific number of steps at the selected speed.

Example Description

This example demonstrates the use of the Stepper 3 click board by driving the motor in both directions for a desired number of steps.

The full application code, and ready to use projects can be installed directly from NECTO Studio Package Manager(recommended), downloaded from our LibStock™ or found on Mikroe github account.

Other Mikroe Libraries used in the example:

- MikroSDK.Board
- MikroSDK.Log
- Click.Stepper3

Additional notes and informations

Depending on the development board you are using, you may need USB UART click, USB UART 2 Click or RS232 Click to connect to your PC, for development systems with no UART to USB interface available on the board. UART terminal is available in all MIKROE compilers.

mikroSDK

This Click board™ is supported with mikroSDK - MIKROE Software Development Kit. To ensure proper operation of mikroSDK compliant Click board™ demo applications, mikroSDK should be downloaded from the LibStock and installed for the compiler you are using.

For more information about mikroSDK, visit the official page.

Specifications

Type	Stepper
Applications	Can be used for the development of unipolar stepper motor control applications. Besides, it can also be used for driving relays, brushed DC motors, etc
On-board modules	ULN2003A - high-voltage high-current Darlington transistor array from Texas

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	Instruments
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Interface	GPIO
Feature	No ClickID
Compatibility	mikroBUS™
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Resources

mikroBUS™

mikroSDK

Learn article: Stepper motors

Click board™ Catalog

Click Boards™

Downloads

Stepper 3 click example on Libstock

UNL2003A datasheet

Stepper 3 click 2D and 3D files v100

Stepper 3 Click schematic v103

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